



Smart conductive nanocomposite hydrogel containing green synthesized nanosilver for use in an eco-friendly strain sensor

Ittipol Taesuwan · Artjima Ounkaew · Manunya Okhawilai · Salim Hiziroglu ·
Wirat Jarernboon · Prinya Chindapasirt · Pornnapa Kasemsiri

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Abstract Eco-friendly smart nanocomposite hydrogels were developed in this study. Green synthesized silver nanoparticles (AgNPs) using extracted sugarcane leaves were incorporated into hydrogel-based polyvinyl alcohol, carboxymethyl cellulose, glycerol, and borax. Different concentrations of silver nitrate at 5, 10, 20, 30, and 50 mM were used to obtain AgNPs (AgNPs-5, AgNPs-10, AgNPs-20, AgNPs-30, and AgNPs-50, respectively). The presence of glycerol and AgNPs in hydrogel improved the long-term

stability of hydrogel. Uniform dispersion of AgNPs-30 in the hydrogel provided the highest tensile strength and strain. After the self-healing test, the hydrogel containing AgNPs were found to have tensile properties comparable to those of conductive hydrogels in previous reports. The conductivity values ranged from 0.14 mS/m to 0.22 S/m. The hydrogel containing AgNPs-30 showed the highest gauge factor (GF) of 5.76 at a stretchability of 265%. The hydrogel containing AgNPs with a low-cost and environment-friendly fabrication process could be applied as a strain sensor.

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I. Taesuwan · A. Ounkaew · P. Kasemsiri (✉)
Sustainable Infrastructure Research and Development
Center and Department of Chemical Engineering, Faculty
of Engineering, Khon Kaen University,
Khon Kaen 40002, Thailand
e-mail: pornkas@kku.ac.th

M. Okhawilai
Metallurgy and Materials Science Research Institute,
Chulalongkorn University, Bangkok 10330, Thailand

M. Okhawilai
Center of Excellence in Responsive Wearable Materials,
Chulalongkorn University, Bangkok 10330, Thailand

S. Hiziroglu
Department of Natural Resource Ecology and
Management, Oklahoma State University, 303-G
Agricultural Hall, Stillwater, OK 74078, USA

W. Jarernboon
Department of Physics, Faculty of Science, Khon Kaen
University, Khon Kaen 40002, Thailand

P. Chindapasirt
Sustainable Infrastructure Research and Development
Center and Department of Civil Engineering, Faculty of
Engineering, Khon Kaen University, Khon Kaen 40002,
Thailand

P. Chindapasirt
Academy of Science, The Royal Society of Thailand,
Dusit, Bangkok 10300, Thailand

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Introduction

Smart hydrogels are three-dimensional polymeric microstructures containing large amounts of water with excellent smart properties, including self-healing, pH-response, and thermal-response (Jiao et al. 2021; Takeno et al. 2020). Among the various types of hydrogel, the conductive hydrogel has attracted much attention because of its use in a variety of potential applications, such as solid electrolytes (Yu et al. 2021), flexible wearable devices, and sensors (Zhu et al. 2021; Xie et al. 2020; Heidarian et al. 2021). Biopolymer, a promising eco-friendly material (Udayakumar et al. 2021), has been widely used for the preparation of polymer-based electrolytes (Duera-mae et al. 2020). Carboxymethyl cellulose (CMC) was used to prepare a conductive hydrogel because of its abundance, low cost, and biodegradability (Uyanga et al. 2021). Blending CMC with a compatible and biodegradable polymer, such as polyvinyl alcohol (PVA), is a practical approach to enhance the stretchability and electrical property of hydrogel (Saadiah et al. 2020; Arefian et al. 2020). Generally, the desirable characteristic of a conductive hydrogel is long-term stability and a wide range of service temperatures, ranging from subzero-low temperatures to high temperatures (Ma et al. 2019). Recently, antifreezing and nondrying hydrogels have been developed to meet this requirement. Cryoprotectant agents, such as ethylene glycol and glycerol, have been incorporated into the hydrogel (Amiryaghoubi et al. 2020). Pan et al. (2018) studied the smart hydrogel-based guar-gum hydrogel/glycerol cross-linked by borax. The obtained smart hydrogel retained its softness and elasticity after freezing at a temperature of $-16\text{ }^{\circ}\text{C}$ for 1 h. The presence of borax creates a dynamic crosslinked network of glycerol-water-borax, which allowed the hydrogel to have ultrafast self-healing ability. The borax incorporated into hydrogel acts not only as a crosslinker but also as a conductivity fluid for a conductive smart hydrogel. However, hydrogel-based polyol and borax usually suffer from poor mechanical properties (Seidi et al. 2020). To improve the properties of the hydrogel,

particularly electrical efficiency, conductive fillers, such as superparamagnetic nanorods (Ferrag et al. 2021), carbon nanotubes (Zhan et al. 2021), graphene (Mehmood et al. 2020), and silver nanowires (Azadi et al. 2020), have been incorporated into hydrogels. Among the various types of conductive fillers, silver is an ideal candidate because of its high conductivity of $6.3 \times 10^7\text{ S/m}$ (Tang et al. 2017), low toxicity to mammalian cells, and facile synthesis via a green process (Mukhejee et al. 2020; Ounkaew et al. 2020). Green synthesized AgNPs can be achieved using plants and their extracts to reduce and convert silver ions into AgNPs (Kumar et al. 2021; Lin et al. 2014; Ibrahim et al. 2021). Wang et al. 2021 reported that the conductivity of polyacrylamide hydrogel remarkably increased with increasing content of green synthesized AgNPs. The GF value and the tensile strength of polyacrylamide hydrogel were determined to be 0.47 and 0.12 MPa, respectively. The obtained AgNPs were reduced by hydroxyethyl cellulose in 24 h. The long reaction time results in high energy consumption and increased cost. To decrease the reaction time of green synthesized AgNPs, the use of a bio-based reducing agent containing polyphenol components was suggested to achieve high reactivity because of their strong radical scavengers (Cheng et al. 2020). Ounkaew et al. (2021) reported that the use of phenolic compounds in extracted spent green tea as a reducing agent for AgNPs can shorten the reaction time approximately six fold as compared to that using starch. Previous investigations carried out studies involving some essential compounds from agricultural and food wastes. For example, caffeic and chlorogenic acids from spent coffee (Batista et al. 2020), catechins, theaflavins, tannins; and flavonoids from spent green tea (Lebaschi et al. 2017); and caffeic and sinapic acids from sugarcane leaves (Abdelwahab et al. 2015) were used as the bio-based reducing agent for the synthesis of AgNPs. Using a bio-based reducing agent represents an alternative to manage and convert wastes to value-added products.

Currently, there is very little information about the use of green synthesized AgNPs from agricultural waste for use in a strain sensor. Therefore, this research aims to develop a conductive hydrogel with multifunctions, viz., antifreezing, nondrying, and self-healing abilities. Smart nanocomposite hydrogels based on CMC/PVA/glycerol/borax containing green synthesized AgNPs from extracted sugarcane leaves

(ex-SGL) were prepared. The effect of silver nitrate (AgNO_3) concentrations on the nanocomposite hydrogel synthesis and properties, i.e., mechanical properties, antifreezing ability, nondrying ability, electrical conductivity, and self-healing ability, were determined. Finally, the sensing ability of the nanocomposite hydrogel to detect human activities was also evaluated.

Experimental

Materials

Sodium CMC (degree of substitution of 0.76, purity of 99.7%, and viscosity of 2,580 mPa·s for 2 wt% solution at 25 °C) purchased from Changshu Wealthy Science and Technology Co., Ltd., China. PVA (average molecule weight 1700–1800) was purchased from Laboratory Reagents & Fine Chemical. Sodium tetraborate or borax (purity of 99.9%) and glycerol (purity of 99.5%) were obtained from Elago Enterprises Pty. Ltd. Silver nitrate (purity of 99.8%) was purchased from RCI Labscan Limited.

Preparation of ex-SGL

The extraction of sugarcane leaves method was modified from (Srikhao et al. 2020). The sugarcane leaves were washed with distilled water and then dried at 90 °C. The dried sugarcane leaves were crushed and then sieved to achieve powder with an average particle size of 150 μm . The ratio of sugarcane leaf powder/distilled water at 1:25 was used to prepare ex-SGL. The sugarcane leaf powder was added into distilled water at a temperature of 90 °C and then stirred for 5 min. The mixture was filtered through Whatman 1.0 filter paper to obtain ex-SGL. The total phenolic content (TPC) of ex-SGL was tested according to the Folin Ciocalteu method. The TPC value was expressed as gallic acid equivalent (GAE). The average value of the TPC from three samples was 162.75 ± 1.34 GAE (mg/L).

Preparation of nanocomposite hydrogel-based CMC/PVA/Glycerol/Borax containing AgNPs

The synthesized AgNPs at the concentrations of AgNO_3 of 0, 5, 10, 20, 30, and 50 mM were denoted

as AgNPs-0, AgNPs-5, AgNPs-10, AgNPs-20, AgNPs-30, and AgNPs-50, respectively. The obtained AgNPs were incorporated into a hydrogel matrix consisting of 2-g CMC, 4-g PVA, 20-ml glycerol, and 1.2-g borax. The preparation of nanocomposite hydrogel is depicted in Supplementary information Fig. S1.

Characterization of conductive nanocomposite hydrogels

The ultraviolet–visible (UV–Vis) spectral analysis of the AgNPs was monitored at 350–510 nm using a UV–Vis spectrophotometer (SPECORD, Analytik Jena, Germany Perkin-Elmer, Lambda 35). X-ray diffraction analysis of conductive nanocomposite hydrogels was carried out using a SmartLab X-ray diffractometer over diffraction angles of $0^\circ \leq 2\theta \leq 80^\circ$; the diffractometer was equipped with a $\text{Cu K}\alpha$ radiation source operated at 45 kV and 200 mA. The interaction between the AgNPs and the polymer matrix was tested using attenuated total reflection Fourier transform infrared (ATR-FTIR) spectroscopy (Jasco 4200). The morphology of the AgNPs and their dispersibility were observed using transmission electron microscopy (Hitachi H-7650). 1 mL of sample solution was diluted 50 times in deionized water, deposited onto 400-mesh carbon-coated Cu grids, and then dried at ambient condition for 24 h. The antifreezing ability of each sample was tested at the freezing point using a Netzsch DSC 214 Polyma differential scanning calorimeter (DSC). All samples were cooled to 20 to -40 °C at a cooling rate of 5°C min^{-1} under 40 mL min^{-1} N_2 as purge gas. The mechanical properties of hydrogel films were tested using a Universal testing machine (Instron, Model 5567) equipped with a 5-kN load cell following the ASTM D 882–10 method (2015). The initial distance between the grips was 50 mm, and a crosshead speed of $5\text{ mm}\cdot\text{s}^{-1}$ was used. The conductivity of each sample was measured via a two probe KEITHLEY model 2400. A voltage in range of 0–12 V was applied in the conductivity tests. The conductivity values (σ) of the hydrogels were calculated from Eq. (1). A Fluke 289 True RMS multimeter was used to detect simultaneous resistance changes under the stimulus.

$$\sigma = \frac{1}{R} \frac{L}{S} \quad (1)$$

where L is the thickness of the hydrogel sample (cm), S is the area of the hydrogel sample (cm²), and R is the resistance of the hydrogel sample (Ω).

Results and discussion

Synthesis and characterization of AgNPs

The formation of AgNPs was investigated using UV–Vis spectroscopy, as shown in Fig. 1. The absorption spectra of the AgNPs were observed in the range of 350–600 nm (Saeb et al. 2014). In this work, absorbance peaks in the range of 425–440 nm were observed. The presence of phenolic compounds acted as bio-based reducing agents and promoted the reduction of silver ions to silver atoms (Srikhao et al. 2021). The blue shift in wavelength of 436 to 432 nm was found for AgNPs-30. The shift of the absorbance peak to shorter wavelength implied a decrease of the AgNPs particle size. AgNPs-30 showed the highest intensity of the absorbance peak at 432 nm. The increase of peak intensity indicated the formation of AgNPs at high concentration (Ounkaew et al. 2021; Affes et al. 2020).

The size and morphology of the AgNPs in the hydrogel were analyzed using TEM, as depicted in

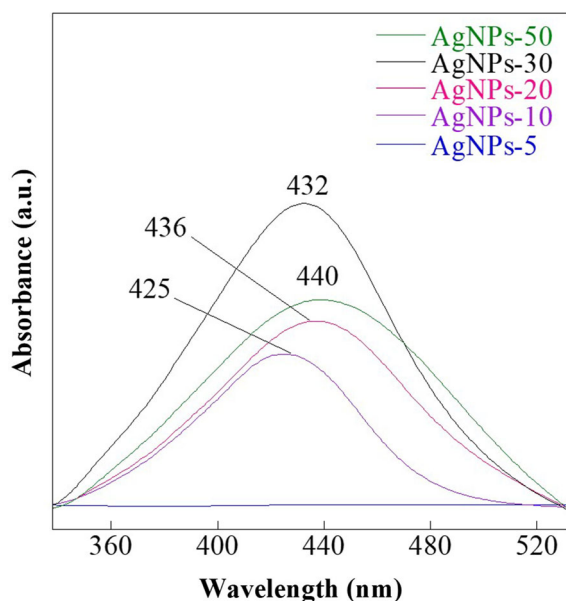


Fig. 1 UV–Vis spectroscopy of hydrogel containing AgNPs

Fig. 2(a). A spherical shape was observed for all hydrogels containing AgNPs. The amount of AgNPs increased with the increase of AgNO₃ content. Rodriguez-Leon et al. (2013) suggested that the kinetic formation and the amount of AgNPs depended on the AgNO₃ concentration. At higher AgNO₃ concentration, a higher yield of AgNP formation could be achieved because more reactant was available for AgNP growth. As shown in Fig. 2(b), the average particle size range of hydrogel containing AgNPs-0, AgNPs-10, AgNPs-20, and AgNPs-30 was 10–45 nm. The increase of particle sizes to 11–55 nm was only found in a hydrogel containing AgNPs-50; this increase was related to the agglomeration of AgNPs. The formation of a large number of AgNPs at high AgNO₃ concentration provided high surface area and surface tension of AgNPs particles, leading to agglomeration (Diniz et al. 2020). The obtained results from TEM analysis were found to be in good agreement with the UV–Vis spectroscopy results.

The crystalline structure of AgNPs was analyzed using XRD. The XRD patterns of green synthesized AgNPs at various concentrations of silver nitrate are illustrated in Fig. 3. The characteristic peaks of AgNPs at 38.05° (111), 46.25° (200), and 66.44° (220) were observed for AgNPs-20, AgNPs-30, and AgNPs-50. The obtained results confirmed the face-centered-cubic lattice of AgNPs (Wang et al. 2021). Note that the small amounts of AgNPs-5 and AgNPs-10 in hydrogels were undetectable by XRD. The mean crystal size of the AgNPs can be calculated from the XRD pattern using the Scherrer equation, as shown in Eq. 2. The average crystallite size (D) values of hydrogel containing AgNPs-20, AgNPs-30, and AgNPs-50 were determined to be 25.24, 18.14, and 30.52 nm, respectively.

$$D = \frac{K\lambda}{(\beta \cos \theta)} \quad (2)$$

where λ is the wavelength of the X-rays, β is the full width at half maximum of the peak, and θ is the peak position.

The ATR-FTIR spectra of hydrogel containing AgNPs are shown in Fig. 4. The spectrum of hydrogel-based CMC/PVA/ex-SGL exhibited the same peaks as those found in neat PVA and neat CMC, as shown in Supplementary information Fig. S2 and summarized

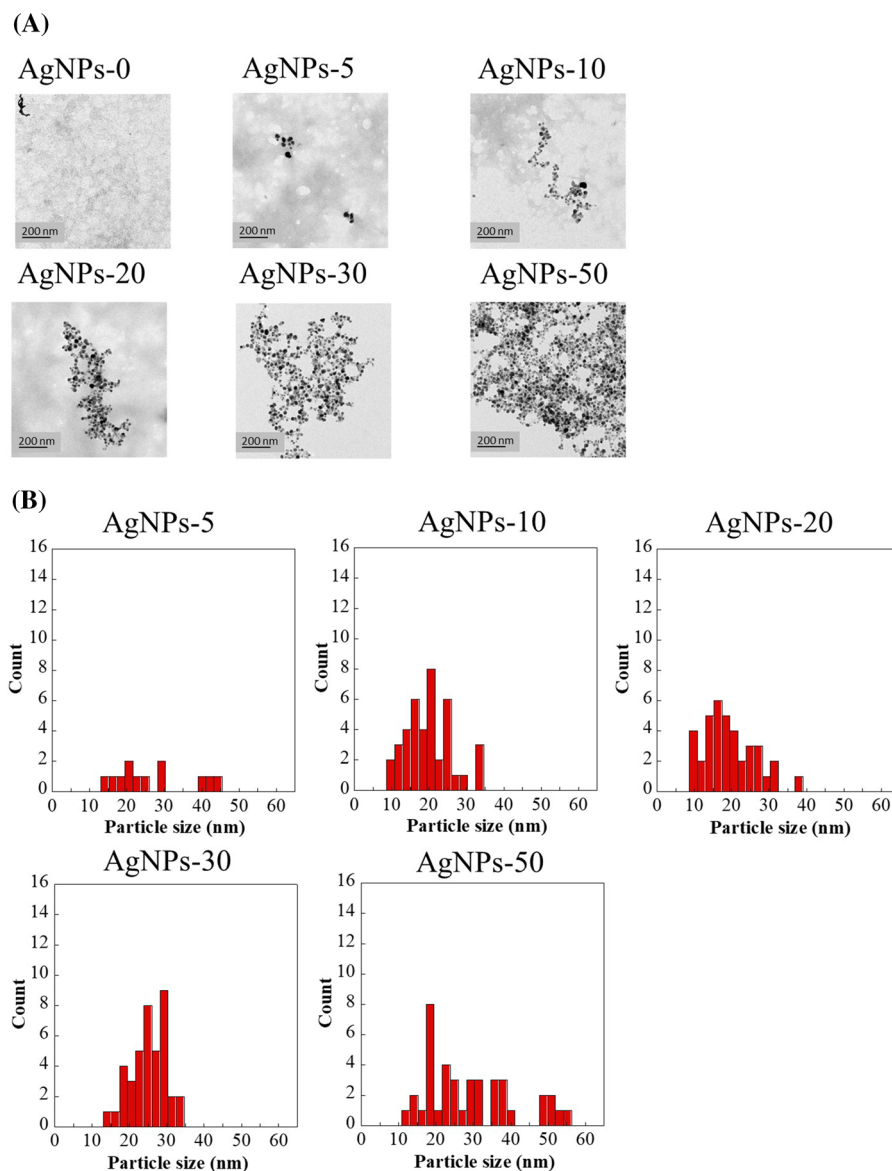


Fig. 2 **a** TEM images of hydrogel containing AgNPs and **b** number size distribution of AgNPs in the nanocomposite hydrogels

in Table S1. The results indicated that no reaction occurred among PVA, CMC, and ex-SGL. Figure 4a shows the ATR-FTIR spectra of hydrogel containing AgNPs-0 and AgNPs-30 in the wavenumber range of 600–1750 cm^{-1} . The absorption peak of B–O–B linkages and the asymmetric stretching vibration of B–O–C linkages in the network were observed at 651 and 816 cm^{-1} , respectively (Sarkar et al. 2019). The sharp peak at 2879 cm^{-1} , which was assigned to C–H

stretching, appeared as shown in Fig. 4b. The broad peak at 3280 cm^{-1} of hydrogel-based CMC/PVA/ex-SGL shifted to 3286 and 3289 cm^{-1} for hydrogel containing AgNPs-0 and AgNPs-30, respectively. This phenomenon indicated the interaction of hydrogen bonding when borax and AgNPs were incorporated into the hydrogel matrix (Dixit et al. 2019).

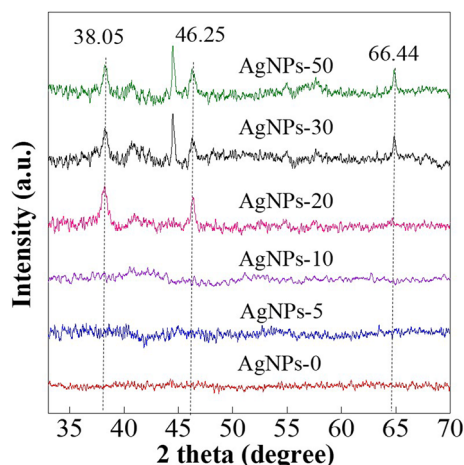


Fig. 3 XRD patterns of hydrogels containing AgNPs

Antifreezing and nondrying abilities of hydrogels containing AgNPs

Conventional hydrogels consist of a large number of free water molecules that freeze at low temperatures or continuously evaporate over a long period of use. The freezing and evaporation of water adversely affect the properties of a hydrogel, especially its mechanical properties (Ma et al. 2019). Figure 5 depicts the mechanical deformation of antifreezing hydrogels containing AgNPs. After freezing at a temperature of $-30\text{ }^{\circ}\text{C}$ for 7 days, all samples could be folded and twisted; whereas, the control samples prepared using only ex-SGL were broken during twisting and folding. Chen et al. (2018) reported that a water-based hydrogel that was frozen via the free water molecules in its structure was rigid and fragile, thereby limiting its applications at low temperature. Fig. S3 shows the DSC thermogram of a control sample. Water crystallization was observed at $2\text{ }^{\circ}\text{C}$. Conversely, the exothermic reaction of water crystallization was absent when glycerol was added into hydrogels containing AgNPs. The large number of hydroxyl groups in glycerol molecules resulted in the interaction with water molecules that could lower the freezing point of water and inhibit water crystallization (Lin et al. 2021). All hydrogels containing AgNPs showed a good antifreezing property for use in cold condition.

The water retaining ability is one of the parameters indicating the long-term stability of hydrogels (Chen

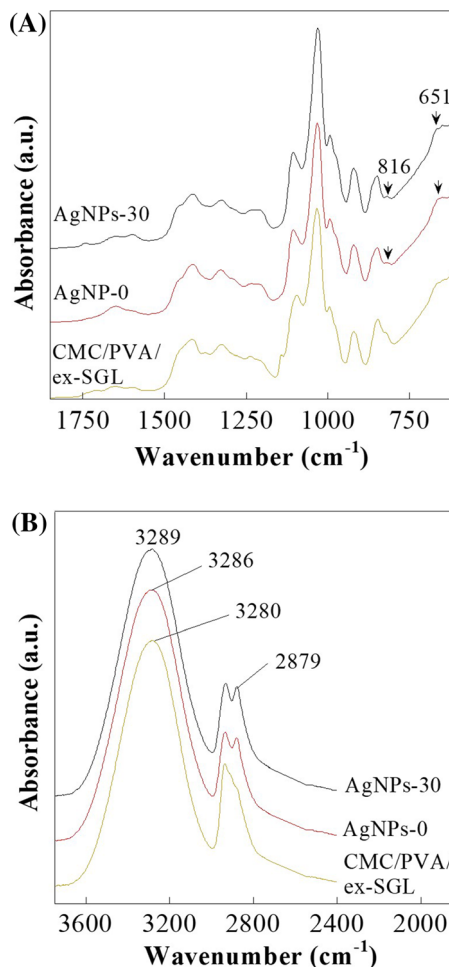


Fig. 4 ATR-FTIR spectra of hydrogel containing AgNPs. **a** wavenumber range of $600\text{--}1750\text{ cm}^{-1}$ and **b** wavenumber range of $2400\text{--}3600\text{ cm}^{-1}$

et al. 2020). The water retaining capacity can be calculated as expressed in Eq. 3.

$$\text{Water retening capacity (\%)} = \frac{W_t}{W_o} \times 100 \quad (3)$$

where W_t and W_o are the instantaneous weight and the original weight of the sample, respectively.

Figure 6 exhibits the water retaining ability of hydrogel containing AgNPs under a temperature of $40\text{ }^{\circ}\text{C}$ and a humidity of 10% for 7 days. The water retaining capacity slightly decreased with increasing storage time. The water retaining capacity was in the range of 91.64%–99.89%, which was comparable with that of other hydrogel containing glycerol at the end of 7 days (Ge et al. 2019). Chen et al. (2018) suggested

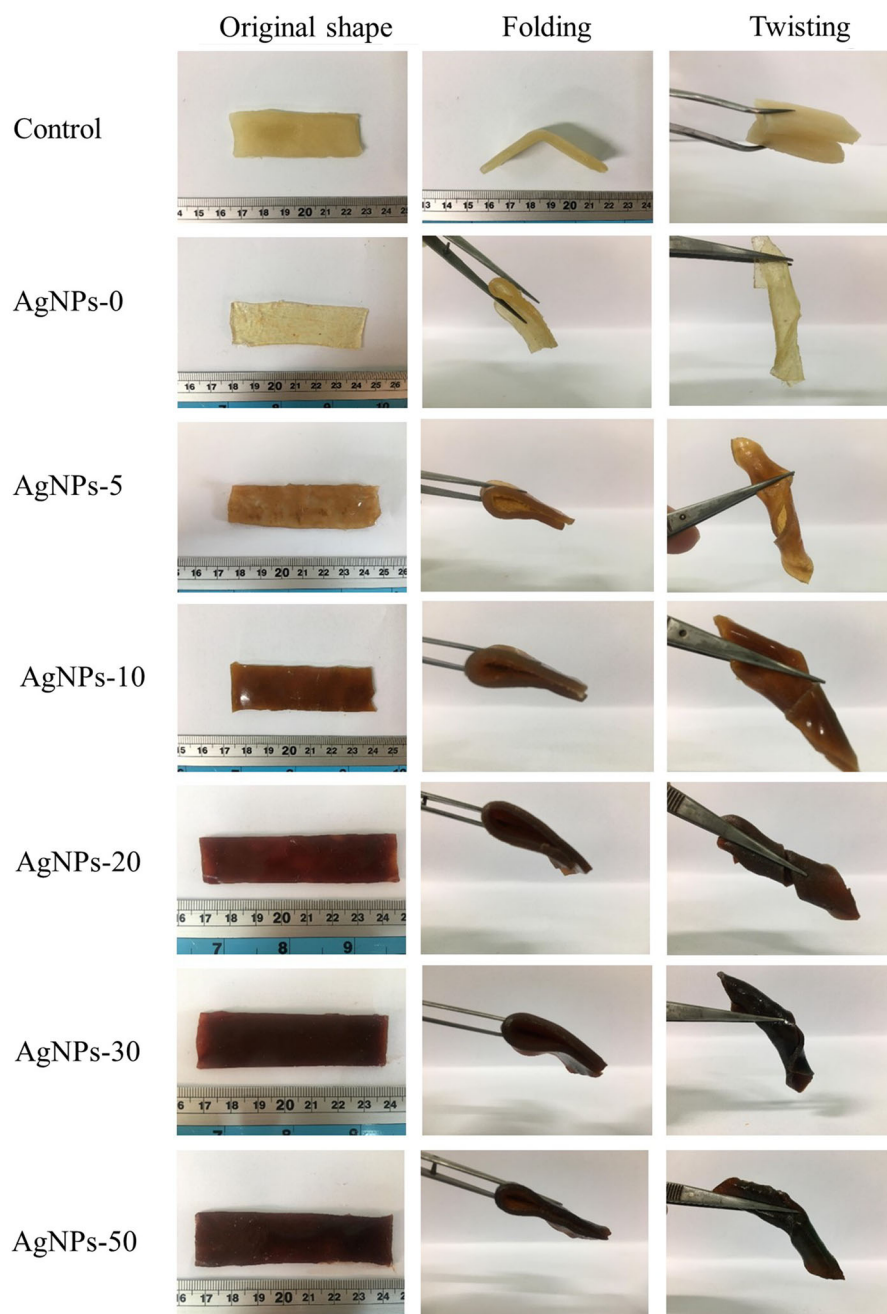


Fig. 5 Mechanical deformation of antifreezing hydrogels containing AgNPs

that the water molecules were captured by glycerol in a hydrogel, thereby impeding water evaporation. Furthermore, the presence of AgNPs in hydrogel also enhanced the water retaining capacity of the samples. The highest water retaining capacity was observed in a

hydrogel containing AgNPs-30. Chen et al. (2020) also observed that the addition of AgNPs into hydrogels provided more compact structures for better moisture retention. The obtained results demonstrated that the antifreezing and nondrying abilities of the

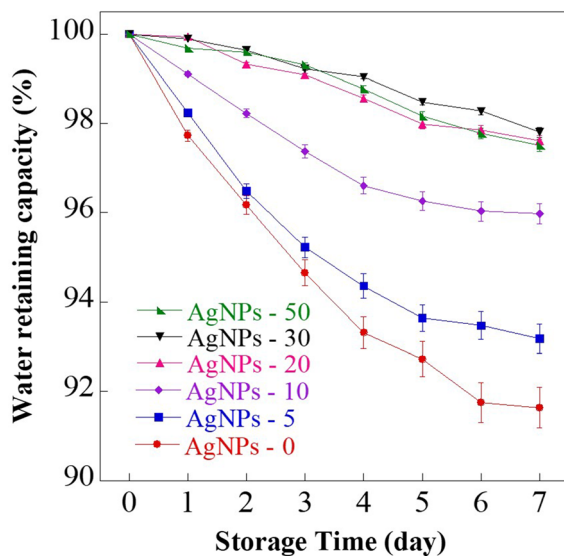


Fig. 6 Water retaining abilities of hydrogels containing AgNPs

hydrogel were effectively enhanced via the addition of glycerol and green synthesized AgNPs.

Mechanical properties and self-healing efficiency of hydrogels containing AgNPs

The mechanical properties and self-healing efficiency of hydrogel samples were evaluated using the tensile test, as illustrated in Figs. 7a–c. The incorporation of AgNPs-30 into hydrogel had a distinct impact on the tensile strength and tensile strain, increasing from 78.68 to 144.86 kPa and from 240 to 265%, respectively. The enhancement in the mechanical properties could be attributed to the reinforcing effect of the nanoparticles with homogenous dispersion. The good dispersion of nanoparticles in a hydrogel can enhance interaction and form many crosslinking points (Zhang et al. 2019; Spoljaric et al. 2014). Conversely, the excess of AgNPs created the aggregates and a denser polymer network, resulting in the decrease of toughness and the stretchability of the hydrogel. The decreases in the tensile strength and tensile strain were observed when AgNPs-50 was incorporated into a hydrogel, as shown in Figs. 7a, b.

Figure 7c illustrates the self-healing efficiency of the hydrogels. The cut hydrogels were allowed to heal at a temperature of 25 °C for 16 h without external stimuli, and then the mechanical properties of the self-

healing samples were tested. The self-healing behavior of hydrogels containing AgNPs resulted from the dynamic and reversible multicomplexation among the hydroxyl groups of the AgNPs, the PVA/CMC copolymer, and borax. The tensile strength and tensile strain of all the hydrogels were lower than those of the original samples. Such results implied that not all of the bonds in a hydrogel were reversible and that some dynamic linkages might be permanently destroyed during damage (Tuncaboylu et al. 2013). The self-healing efficiency decreased from 80.90% for the hydrogel containing AgNPs-0 to 38.59% for the hydrogel containing AgNPs-50. This observation is consistent with the previously published works. It was also reported from the past studies that the self-healing ability of hydrogels could be enhanced by decreasing their crosslinking density (Spoljaric et al. 2014). At low crosslinking density, the polymer chains had greater flexibility and movement ability, leading to an increase in bond reformation across the broken interface. For a hydrogel containing AgNPs, the presence of AgNPs in the hydrogel network might hinder the polymer chain mobility, resulting in a decrease of the self-healing efficiency. Note that the healed hydrogel containing AgNPs had tensile strength and tensile strain comparable to those of hydrogel composites previously reported, such as PVA/borax containing nanofibrillated cellulose (Spoljaric et al. 2014) and hydroxypropyl guar gum/graphene oxide (Sun et al. 2020).

Conductivity of hydrogels containing AgNPs

The electrical conductivity values of the samples are depicted in Fig. 8(A). The conductivity of AgNPs-0 was 0.14 mS/m. The crosslinking mechanism of borax in the hydrogel-based CMC, PVA, and glycerol created borate ions that acted as a polyelectrolyte (Pan et al. 2018). The presence of green synthesized AgNPs in hydrogels remarkably increased the conductivity. The AgNPs in the hydrogel served as the electrical pathway. At high AgNPs contents, the interconnected nanoparticles created a network structure, thereby increasing the conductivity (Huang et al. 2021). This result is in good agreement with TEM images, as discussed in the morphology analysis. The values of conductivity of hydrogel containing green synthesized AgNPs ranged from 0.22 mS/m for AgNPs-5 to 0.22 S/m for AgNPs-50. The conductivity

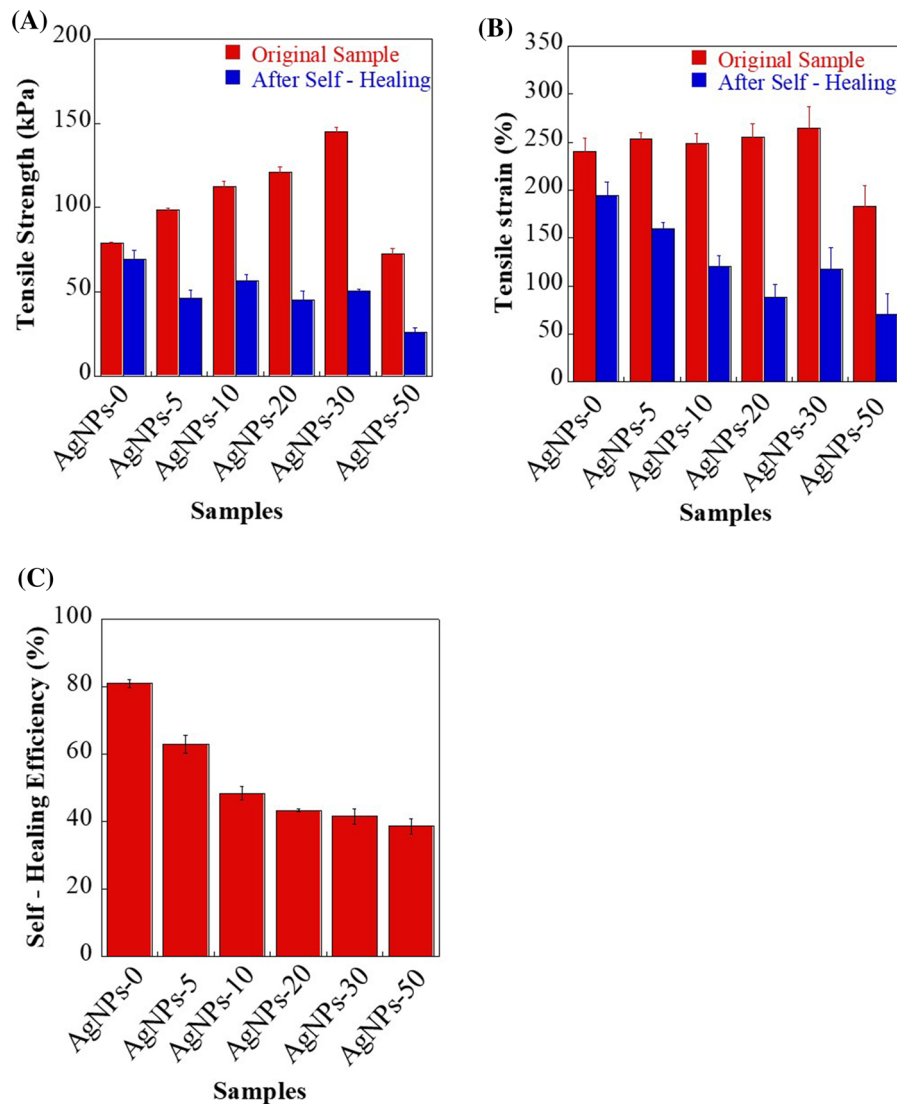


Fig. 7 Mechanical properties and self-healing efficiency of hydrogels containing AgNPs. **a** Tensile strength, **b** tensile strain, and **c** self-healing efficiency

values of hydrogels containing green synthesized AgNPs were comparable with those of other reported composite hydrogels, viz., polyacrylamide hydrogel containing decorated hydroxyethyl cellulose with AgNPs (0.08–0.32 S/m) (Wang et al. 2021), PVA/borax hydrogel containing decorated cellulose nanocrystal with tannic and silver (0.16–4.61 S/m) (Lin et al. 2019), and thin film hydrogel containing titanium dioxide (0.11 S/m) (Ryplida et al. 2019).

GF is a parameter representing the response of a sensor to external stimuli. A high GF value implies

good accuracy of the sensor to detect external changes (Han et al. 2020). The GF values of hydrogel containing AgNPs were calculated at strain values of 0%–265%, as shown in Fig. 8(b). The GF of hydrogel containing AgNPs-0 was 2.09. The presence of AgNPs in hydrogel increased the GF value because of the electrical pathway of the AgNPs. A similar observation was reported by Jing et al. (2019), who observed that the addition of silver nanowires into a gelatin hydrogel can improve the GF value by approximately two fold. The highest GF value of 5.76 was observed

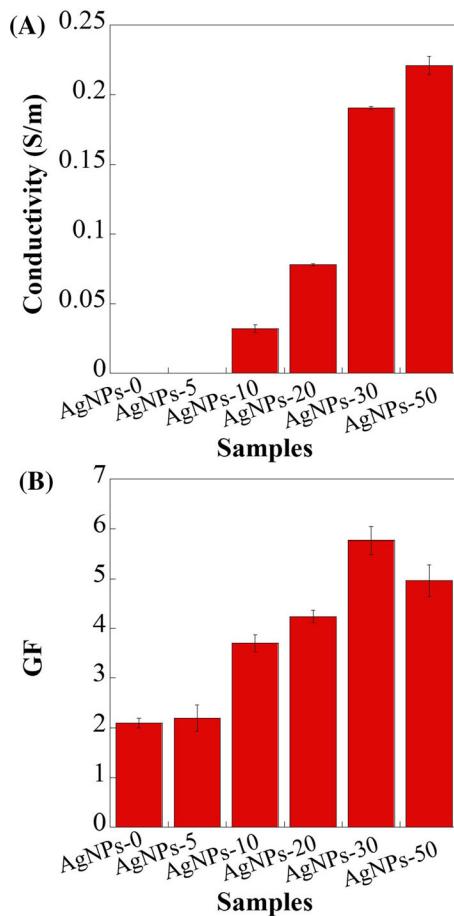


Fig. 8 Electrical conductivities of the hydrogel samples. **a** Conductivity and **b** GF values

in a hydrogel containing AgNPs-30. The obtained hydrogel containing green synthesized AgNPs-30 that integrated self-healing, antifreezing, and nondrying abilities had a sensitivity and detection range

comparable to those of other reported hydrogels in the previous studies, as summarized in Table 1.

Application of hydrogels containing AgNPs for monitoring human motion

The hydrogel containing AgNPs-30 was used as a strain sensor to detect human motions, as depicted in Fig. 9. The hydrogel containing AgNPs-30 can detect the motions of a finger, a wrist, an elbow, and a knee. The signal of motion was achieved from the variation of the current produced by the sensor. The minimum current was created when the sensor was in the relaxation state. The current signal increased and reached the highest value when the attached sensor on a body was bent (Zhao et al. 2020). The obtained signals can reflect each pattern of motion with a stable baseline. This behavior implied a good recoverability of the sensor during the monitoring process (Peng et al. 2020). The reproducibility and stability of the signal to repeated bending and releasing motions were also observed. Furthermore, the hydrogel containing AgNPs-30 was also used to detect human motion under subzero environment ($-30\text{ }^{\circ}\text{C}$) and high temperature ($40\text{ }^{\circ}\text{C}$), as shown in Fig. 10. The original sample and the self-healing sample can accurately detect the motion with repeatable cycling stability at temperatures of $-30\text{ }^{\circ}\text{C}$ and $40\text{ }^{\circ}\text{C}$. Note that the hydrogel containing AgNPs-30 could be developed as a highly reliable strain sensor for detecting human motions.

Conclusions

In summary, smart hydrogel nanocomposites containing AgNPs were successfully developed as strain

Table 1 Comparative study of strain sensors

Strain sensor based hydrogel	GF	Stretchability (%)	References
Polyvinylpyrrolidone/PVA/ Cellulose nanocrystals	0.48	200	Liu et al. (2017)
α -liopic acid/acrylic acid-ionic	1.46–3.69	50–300	Dang et al. (2019)
Chitosan/carbon nanotubes	4	10	Wu et al. (2018)
Gelatin/silver nanowires	1.1	200	Jing et al. (2019)
Hydrogel containing AgNPs-30	5.76 ± 0.28	265	This study

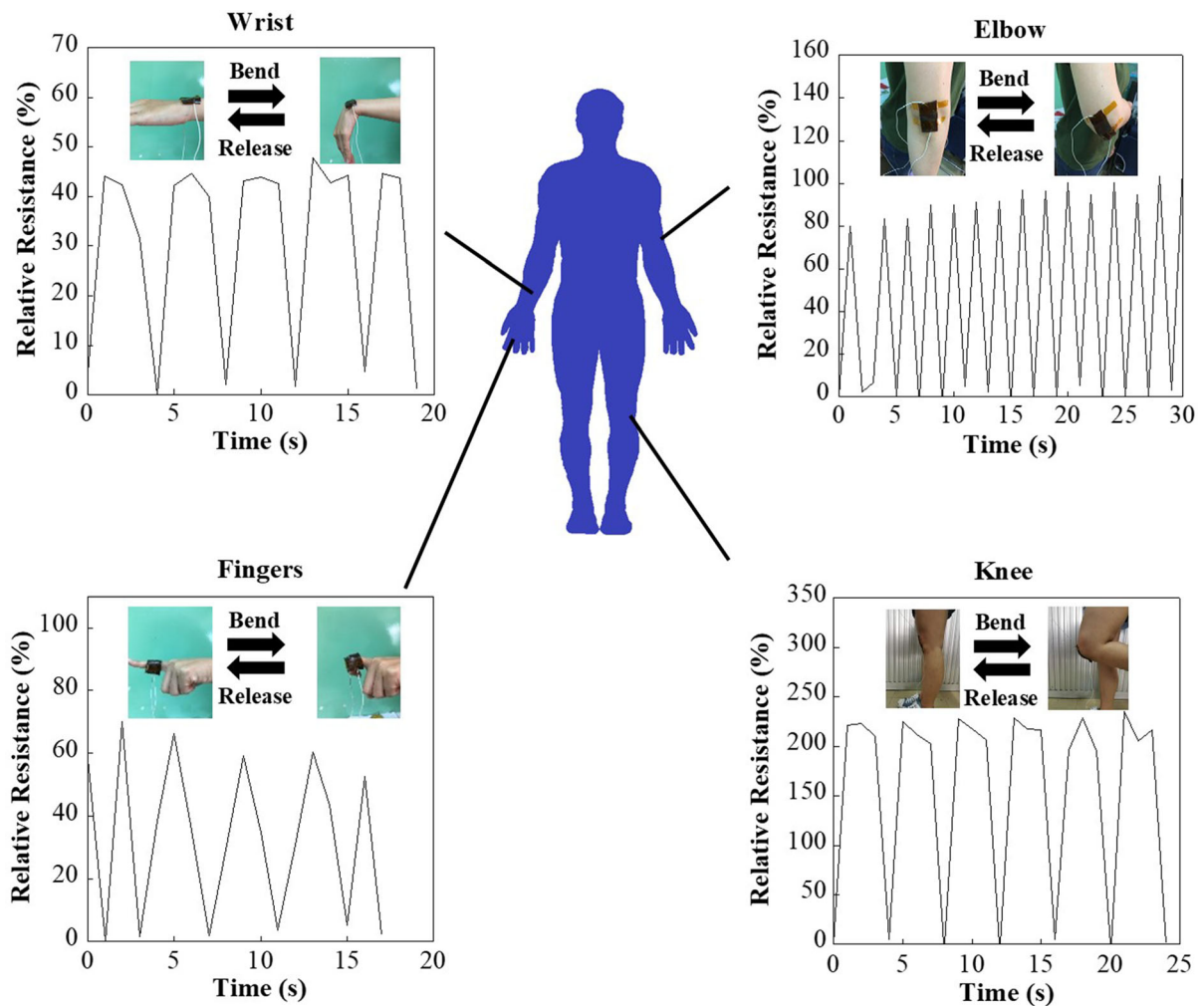


Fig. 9 Resistance change of the hydrogel containing AgNPs-30 on a body upon the motion of the wrist, elbow, finger, and knee

sensors. The green synthesized AgNPs using ex-SGL were characterized using UV–Vis, XRD, and TEM to determine their structure and morphology. The incorporation of glycerol and AgNPs into the hydrogels remarkably enhanced the antifreezing and nondrying abilities at $-30\text{ }^{\circ}\text{C}$ and $40\text{ }^{\circ}\text{C}$, respectively, for 7 days. The uniform dispersion of AgNP-30 in hydrogel increased the tensile stress and strain up to 88.91% and 10.42%, respectively. The self-healing efficiency decreased from 80.90 to 38.59% when AgNPs-50 was incorporated into hydrogel because of

the hindrance effect of AgNPs on the polymer chain mobility. The mechanical properties of healed hydrogel containing AgNPs were found to be comparable to those of the other conductive hydrogels in previous reports. The conductivity of hydrogel substantially increased with the increase of the AgNP content from AgNPs-5 to AgNPs-50. The highest GF at stretchability of 265% was found in the hydrogel containing AgNPs-30. The obtained hydrogel nanocomposites displayed good flexibility and responded well to multiple strain and human motions.

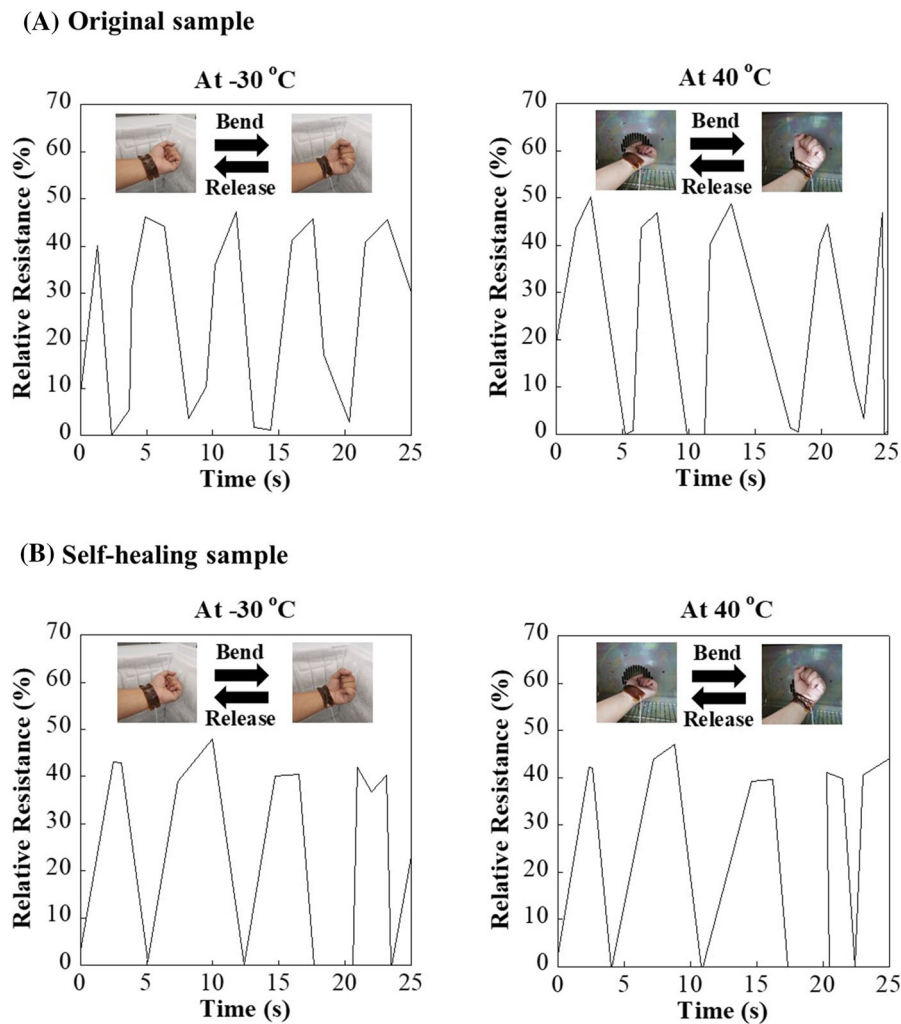


Fig. 10 Resistance change of the hydrogel containing AgNPs-30 upon wrist motion at — 30 °C and 40 °C **a** original and **b** self-healing samples

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Declarations

Conflict of interests The authors declare no competing interests.

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